Web Application Frameworks
CS457
Fall 2025

Task # 6 CLO-2

Deadline: 24th October 2025 before 15:00

Note: This is an individual assignment. Name your zip file as WAF-T6-YourName.zip/rar. Submit your assignment in fs folder "WAF- F25-T6".

▪ Middlewares

You're going to build a Task Manager API project that manages tasks, stores data in files, and uses middleware for rate limiting and logging. Follow these steps to make sure everything is clear and organized:

1. Task Manager API
   - Each task could include details like a title, description, status, and creation date.
2. Storage
   - Use a local and remote database to store your tasks.
   - Choose a simple format (like JSON) so you can easily read from and write to the file.
3. Rate Limiting Middleware
   - Create middleware to monitor and limit the number of API requests coming from a
   - single IP address.
   - Decide on a safe request limit per minute (or another time period) to prevent abuse.
   - If someone exceeds the limit, return an appropriate error response.
4. Logging Middleware
   - Build another middleware component that logs detailed information about every
   - request.
   - Log data like the IP address, endpoint accessed, method (GET, POST, etc.), and
   - timestamp.
   - Ensure these logs are stored or printed in a way that you can refer back to them.